Onur Kansoy

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Research interests

Multi-agent reinforcement learning, emergent behaviour, social dilemmas, opponent modelling, game RL.

Education

B.Sc. in Computer Engineering, Izmir Institute of Technology Sep 2021 to Jan 2027 (expected)

CGPA 3.10 / 4.00. Term GPA 3.25 to 3.40 across the last four completed terms.

Research

Smart Zebra: A Prompt-to-GRPO Recipe for Logic-Grid Reasoning in Small Language Models 2025 to 2026

First author, with Emre Ertürk. Advisor: Assoc. Prof. Selma Tekir. Senior thesis, funded by TÜBİTAK 2209-A. Manuscript in revision. SFT and GRPO pipeline on Qwen2.5-3B-Instruct, and a reimplementation of the ZebraLogic generator with Z3.

Hunter-Gatherer Survival Oct 2026 to present

Custom finite MDP and Gymnasium environment for CENG567, with multi-agent experiments in progress.

Hide and Seek Sep 2026, paused

1v1 continuous self-play environment with custom recurrent PPO and an asymmetric critic, 56.6M training steps.

Grants

TÜBİTAK 2209-A University Students Research Projects Support Programme, for Smart Zebra.

Experience

Undergraduate Researcher, Digital Humanities and Artificial Intelligence Laboratory, IZTECH Feb 2026 to present

Annotated and classified 257 Ottoman Turkish documents with an AI-assisted workflow, and built a ground-truth dataset for fine-tuning domain-specific language models.

Website and Localization Specialist, Department of General Cultural Courses, IZTECH Nov 2025 to Mar 2026

Audited the department website, improved its structure, and closed the content gaps between its Turkish and English pages.

AI Engineering Intern, Research Ecosystems, Teknopark Izmir Jul to Sep 2025

Turned a static chatbot workflow over a research database into a multi-agent system with LangGraph and PostgreSQL, tested it on bilingual question sets, then built GPT-Deep-Search for finding academic profiles.

Game Developer and Designer, Teknopark Izmir Nov 2021 to Feb 2023

Joined after our 7-person team of preparatory-class students won 2nd prize at the Kuleizmir Game Hackathon with Smyrna Zindanları, and worked on game prototypes.

Selected coursework

Reinforcement Learning (CENG567, in progress), Artificial Intelligence (CENG461, in progress), Introduction to Computer Graphics (CENG487, in progress), Introduction to Machine Learning (AA), Natural Language Understanding and Generation (BA), Analysis and Design of Algorithms (AA), Senior Thesis and Seminar I, II (AA, AA), Stochastic Processes (DC), Probability and Statistics (CB), Human Computer Interaction (BB).

Skills

Languages: Python, JavaScript/TypeScript, GDScript, SQL. **ML:** PyTorch, Gymnasium, PettingZoo, TRL, PEFT, Unsloth, vLLM. **Agents:** LangGraph, LangChain, OpenAI Agents SDK. **Systems:** FastAPI, PostgreSQL, pgvector, Docker, Linux. **Other:** Godot, React Native, LaTeX.

Games

Crossblocks (Google Play), Bounce In (testing), Circle In (in development). 2nd prize, Kuleizmir Game Hackathon (2021). Jamination 5 and HoBI'22 game jams (2022). IZTECH DuckJam (2025). Hugging Face Build Small Hackathon (2026).

Leadership

Iztech Bridge Society, former president and current board member. Iztech Neuroscience Society, former board member.

Languages

Turkish (native), English (B2), Italian (beginner).